# PREMIXED INSULIN 70/30

High-alert drug
Fixed meal timing

**CHEMICAL NAME** 70% isophane (NPH) human insulin + 30% soluble human insulin (or 70% insulin aspart protamine + 30% aspart)

**CLASS** Biphasic (premixed) insulin

**TRADE NAMES** Mixtard 30 · Humulin M3 / 70/30 · NovoMix 30 (analogue) · Humalog Mix25/50

## THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes needing basal and prandial cover with fewer injections.
- Type 1 diabetes in patients unable to manage basal-bolus regimen.
- Regular meal patterns.

## PHARMACODYNAMICS

- Soluble component covers the meal; NPH component provides background insulin.

## ADMINISTRATION

- **SC** twice daily: 30 min before breakfast and evening meal (human mix) or immediately before meals (analogue mix).
- Roll/invert to resuspend (cloudy).
- Never IV.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Weight gain.
- Injection-site reactions, lipohypertrophy.

### ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia (late morning, night) if meals missed or delayed.
- Hypokalaemia.

## DOSAGES

- **Dose:** individualised, usually twice daily before breakfast and evening meal.
- Adjust to pre-meal glucose readings per diabetes team.
- Inject about 30–45 min before the meal (human premix).

## CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia; IV use; hypersensitivity.

### KEY INTERACTIONS

- As for other insulins.

## PHARMACOKINETICS

- **Onset:** 30 min (human) / 10–20 min (analogue). **Peak:** dual — 2–4 h and 4–8 h. **Duration:** up to 24 h.

## NURSING MANAGEMENT

- Coordinate injection with meal times; ensure meals not delayed.
- Monitor pre-meal and bedtime glucose.
- Resuspend before each dose.
- Withhold/adjust if patient nil by mouth; seek advice.

## PATIENT EDUCATION

- Inject before breakfast and evening meal; do not skip meals.
- Keep regular meal times and carbohydrate amounts.
- Carry glucose for hypos.
- Roll the pen 10 times before injecting.

**KEY POINTS**

- 70% intermediate + 30% short-acting.
- Requires regular meals.
- Twice daily before breakfast and dinner.
- Hypoglycaemia if meals missed.

**SOURCES CHECKED** **[590]** HUMULIN M3 (Mixture 3) 100IU/ml suspension for injection in cartridge (UK SmPC)

*Full references p. 432–457*